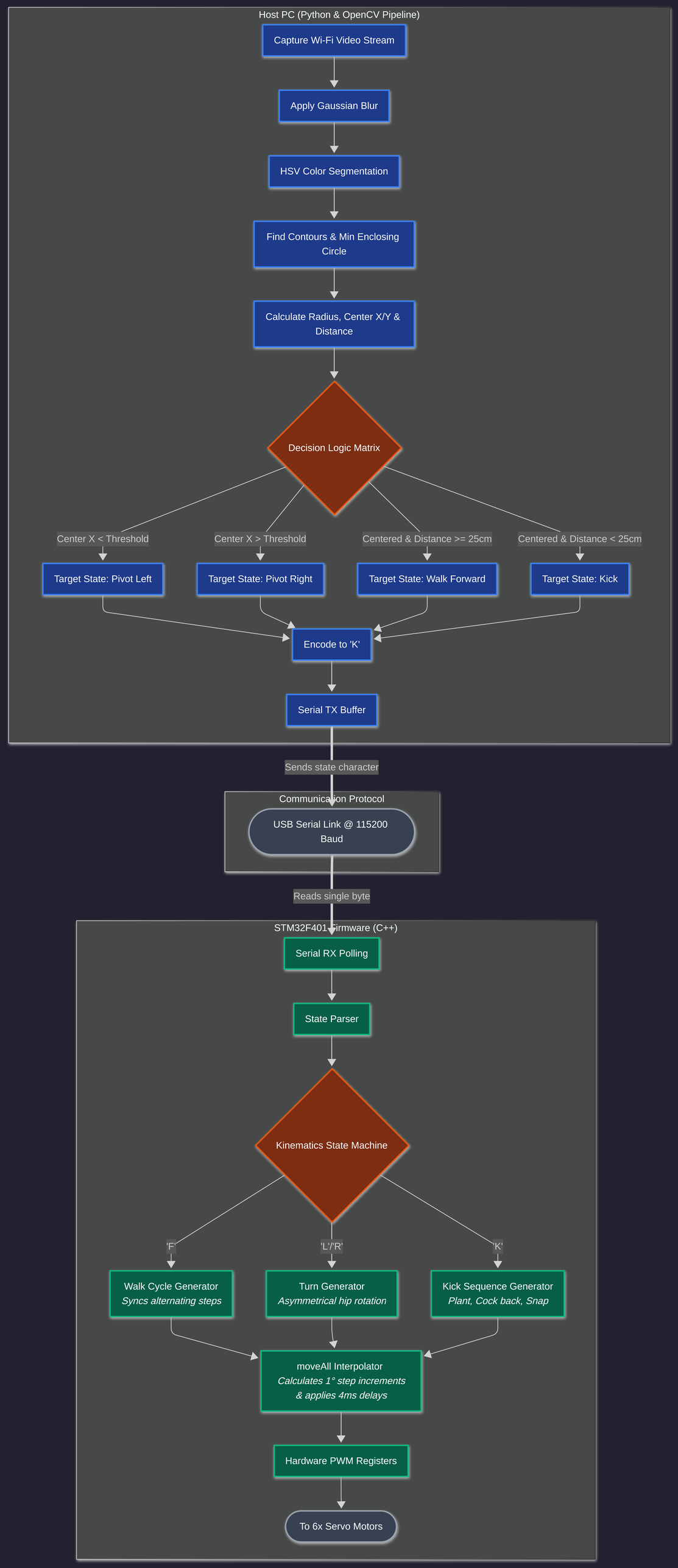
Host PC (Python & OpenCV Pipeline)
Capture Wi-Fi Video Stream
Apply Gaussian Blur
HSV Color Segmentation
Find Contours & Min Enclosing Circle
Calculate Radius, Center X/Y & Distance
Decision Logic Matrix
Center X < Threshold
Center X > Threshold
Centered & Distance >= 25cm
Centered & Distance < 25cm
Target State: Pivot Left
Target State: Pivot Right
Target State: Walk Forward
Target State: Kick
Encode to 'K'
Serial TX Buffer
Sends state character
Communication Protocol
USB Serial Link @ 115200 Baud
Reads single byte
STM32F401 Firmware (C++)
Serial RX Polling
State Parser
Kinematics State Machine
'F'
'L'/'R'
'K'
Walk Cycle Generator
Syncs alternating steps
Turn Generator
Asymmetrical hip rotation
Kick Sequence Generator
Plant, Cock back, Snap
moveAll Interpolator
Calculates 1° step increments & applies 4ms delays
Hardware PWM Registers
To 6x Servo Motors